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Effects of exercise during pregnancy on maternal and newborn outcomes

Yu Shen[1], Yi Wang[2] and Peng Huang[1*]

Abstract
There is some debate over whether exercising during pregnancy has any impact on the health of the mother or baby. While most studies indicate that moderate exercise during pregnancy is beneficial for mothers and does not harm newborns, some studies suggest that it may divert oxygen and nutrients away from the baby, which could negatively affect development. However, due to variations in the type and amount of exercise, as well as different measures used, it is difficult to compare studies. Nevertheless, most studies on the relationship between exercise and pregnancy have not reported the effects of exercise on maternal or fetal health. This review concludes that moderate exercise can benefit both mothers and babies and recommends that uncomplicated mothers exercise at a moderate level throughout their pregnancy without risk to maternal or fetal health.

Keywords Exercise during pregnancy, Maternal, Newborn outcome

Specificity of pregnancy and childbirth

Pregnancy is a special period during which the maternal body undergoes dramatic changes to accommodate the healthy growth of the fetus in the body coupled with the stresses associated with the growth of the fetus (as shown in Fig. 1), and these changes, in turn, may have an impact on the long-term health of the fetus [1].

Shortly after conception, the maternal cardiovascular system undergoes dramatic changes that last for at least a year after delivery [2]. During pregnancy, plasma volume increases by approximately 40–50% during gestation, with a subsequent increase in cardiac output of 35–45% and an increase in heart rate until full term [3–5]. Moreover, total peripheral vascular resistance decreases dramatically during pregnancy [4, 6]. In summary, cardiac output increases considerably during pregnancy and is redistributed to favor flow to the uterus, breasts, and kidneys [7, 8].

During pregnancy, maternal metabolism is altered, with a 20% increase in the basal metabolic rate by late pregnancy [9]. During early pregnancy, blood glucose levels begin to rise in pregnant women due to hormonal changes and to meet the nutritional needs of the fetus [10]. During mid and late pregnancy, in preparation for labor and breastfeeding, pregnant women's fat stores increase [11]. In late pregnancy, calcium and phosphorus levels in the blood of pregnant women increase to meet the needs of fetal bone development [12] Due to the rapid growth of the fetus and the placenta, there is a substantial increase in maternal weight during mid- to late pregnancy, and the Institute of Medicine guidelines in 2009 recommended an increase in gestational weight of 12.5–18, 11.5–16, 71-11.5.5, and 5–9 kg for underweight, normal weight, overweight, and obese women, respectively [13].

As the fetus grows, the diaphragm rises 4 cm, the chest cavity expands, and the ribs rise to mitigate the change

*Correspondence:
Peng Huang
huangp@bsu.edu.cn
[1]School of Sports Medicine and Rehabilitation, Beijing Sport University, Beijing 100084, China
[2]Department of Physical Education, Renmin University of China, Beijing 100872, China

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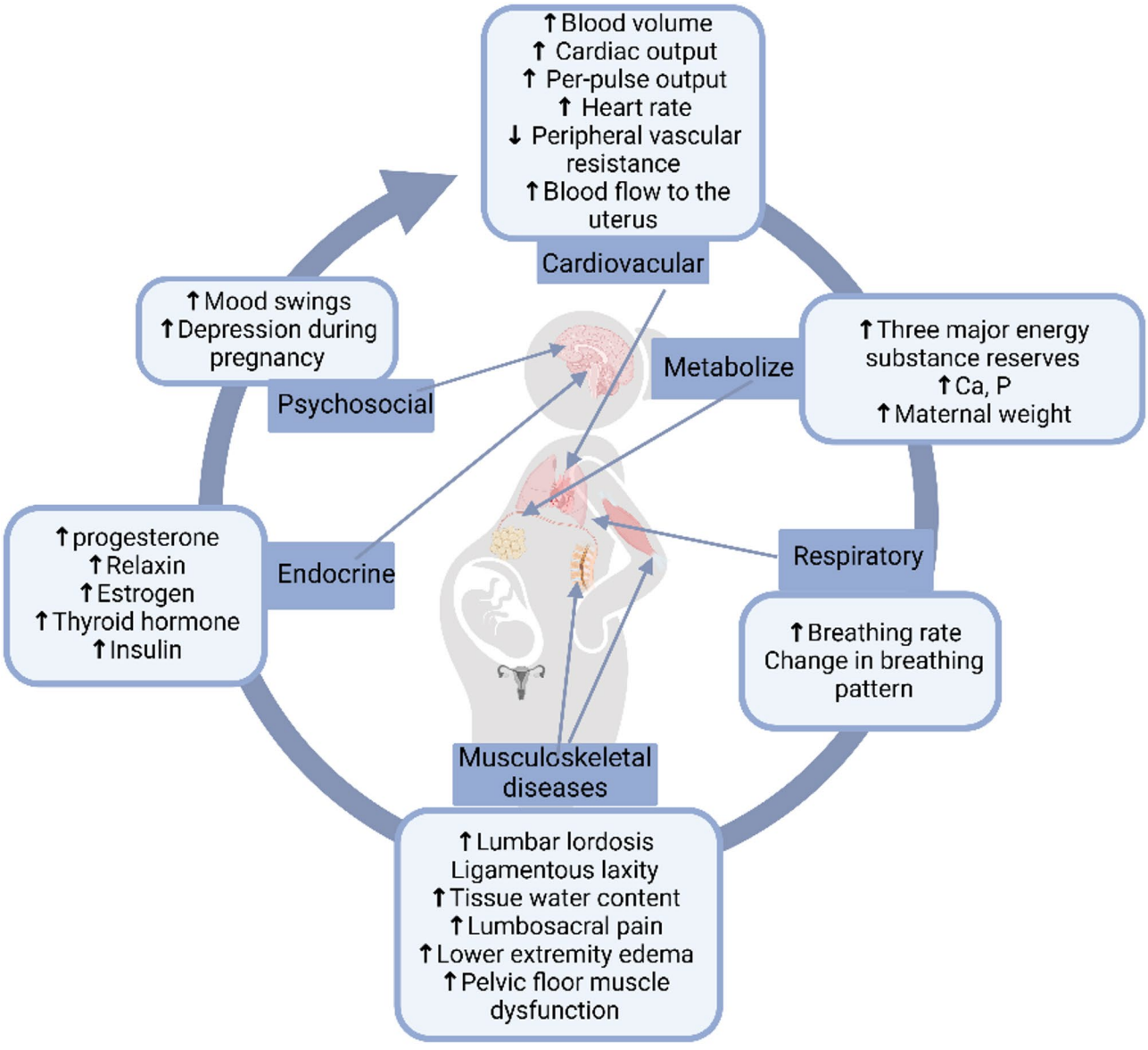

Fig. 1 Impact of pregnancy on maternity

in the lung space associated with diaphragmatic elevation [14–16]. A decrease in lung volume leads to an increase in the respiratory rate to meet the body's demand for oxygen [17]. However, lung volume may still decrease, which will cause pregnant women to feel short of breath, especially in late pregnancy [18].

During pregnancy, the mother undergoes several hormonal changes to support pregnancy and fetal development, and these hormones are mainly secreted by the corpus luteum, placenta, and embryo [19–21]. For example, in early pregnancy, there is an increase in the secretion of estradiol, progesterone, cortisol, relaxin, and insulin [19, 22]; and in late pregnancy, there is an increase in the secretion of lactogen, thyroid hormones, and prolactin [12, 23]. These hormonal changes will have an impact on maternal cardiovascular function, respiration, the musculoskeletal system, and mental status [7, 8, 18, 24, 25].

During pregnancy, women may develop various musculoskeletal disorders. The lumbar spine is more anteriorly convex due to fetal, uterine, and breast growth [26], ligamentous laxity due to relaxin secretion, and fluid retention in connective tissues due to estrogen which predisposes patients to lumbosacral pain [25]. Increased blood volume, altered venous smooth muscle tone, increased intraventricular pressure due to decreased venous return from the lower body by the pregnant uterus, decreased plasma colloid osmolality and genetic susceptibility to lower extremity edema due to decreased serum protein and albumin levels in late pregnancy [27]

lead to pain, heaviness, nocturnal cramps and sensory abnormalities in women [28]. In addition, pregnancy and childbirth are considered risk factors for PFD. The risk of urinary incontinence increases with the progression of pregnancy due to altered hormonal status and increased weight of the uterus on the pelvic floor [29, 30].

During this period, women may experience mood changes. Studies have shown that mood during pregnancy follows a "U" curve, with negative, distressed, and irritable moods being more common in the early and late stages of pregnancy and less common in the middle of pregnancy [31]. In severe cases, they may even lead to depression. Nearly one in five women of childbearing age will suffer from depression at some point in their lives, and pregnancy is a period of particular vulnerability to depression compared to other periods in a woman's life [32]. Postpartum depression is common, but one study reported that the prevalence of depression during pregnancy was 20.7% [33], compared to only 9.1% at 8 weeks postpartum [34].

Exercise and pregnance

With the popularization of 'exercise is medicine', many pregnant women are also consulting clinicians about exercise training programs. Advice on exercise during pregnancy has long been based more on social and cultural beliefs or 'common sense' than on any hard scientific evidence. It has been assumed that mothers may experience bone loss, and a shift in the center of gravity, and are prone to a variety of injuries during exercise, such as miscarriage after a fall. However, in reality, the benefits gained from exercise far outweigh the risks. In the past, exercise has been shown to have many benefits for pregnant women without the risk of adverse effects on maternal and infant health. Vermani et al. [35] reviewed worldwide public guidelines for physical activity during pregnancy and reported that physical activity during pregnancy is beneficial for both the mother and her unborn child. However, controversy still surrounds how exercise affects pregnancy complications; and fetal health. Therefore, the following section describes the effects of exercise on maternal complications and fetal birth outcomes.

Effects of exercise on maternal and fetal health during pregnancy (as shown in Fig. 2)

Fetal growth and development

Through the umbilical cord and placenta, oxygen and nutrients are transported to the fetus, and metabolites such as carbon dioxide are subsequently transported. Changes in the umbilical cord and placenta are closely related to fetal growth and development. Exercise during pregnancy has been found to produce many favorable changes in the placenta [36–39]. A meta-analysis by Kubler et al. [40] revealed an increase in placental parenchymal tissue, which manifests itself as an increase in chorionic tissue, in addition to the fact that a reduction in placental oxidative stress during exercise perhaps improves placental vascularization [39, 41], All of these factors are favorable for gas exchange between the mother and fetus in terms of oxygen diffusion capacity. Neonates of mildly and moderately active women have higher cord blood HDL cholesterol levels than neonates of inactive women [42].

In addition, fetal indicators also reflect the effect of exercise during pregnancy on the fetus. Prenatal exercise improves fetal cardiac function and pumping parameters [43] prevent excessive neonatal fat accumulation, and maintain fetal muscle mass [44]. A 2020 study also showed that prenatal participation in recommended levels of aerobic exercise improves fetal cardiac function and outflow parameters [43]. In addition, aerobic exercise increases thyroid levels in the maternal circulation and in the embryo, enhances embryonic myogenesis, and favors fetal neurodevelopment [45]. Compared with active mothers, Three-month-old infants of inactive mothers have a greater chance of having a low tPTEF/tE (ratio of peak expiratory tidal flow to expiratory time, tPTEF/tE) of < 0.25 [46].In a multicenter RCT published in 2021 with a 7-year follow-up, after mothers were randomized to receive either a structured exercise programme for 12 weeks during pregnancy or standard antenatal care, a survey revealed an association between children's leisure recreation time and intensity and mothers' daily physical activity and exercise intensity [47].

Cardiovascular

Although the fetal heart rate increases during exercise, the mother and fetus can adapt well to exercise. Sene et al. [48] investigated the effects of exercise before or during pregnancy on maternal blood pressure and reproductive outcomes as well as fetal development in spontaneously hypertensive rats (SHRs). The rats swam from day 0 to day 20 of gestation, and measurements on day 21 revealed a reduction in blood pressure. A study by Collings et al. [49] revealed that more vigorous physical activity in pregnant women increased the cardiac vagal pressure reflex, which facilitates blood pressure control during pregnancy. In an RCT involving 66 pregnant women, Makaruk et al. [50]reported that a significant reduction in maternal heart rate was achieved after performing supervised physical activity of moderate intensity, three times per week, for 50–60 min/session, and lasting from 13 to 40/41 weeks of pregnancy. In a recent prospective RCT involving 32 nulliparous women, Osman et al. [51] investigated the effects of prepregnancy cardiovascular monitoring, exercise, and beetroot juice on cardiovascular parameters in women with planned

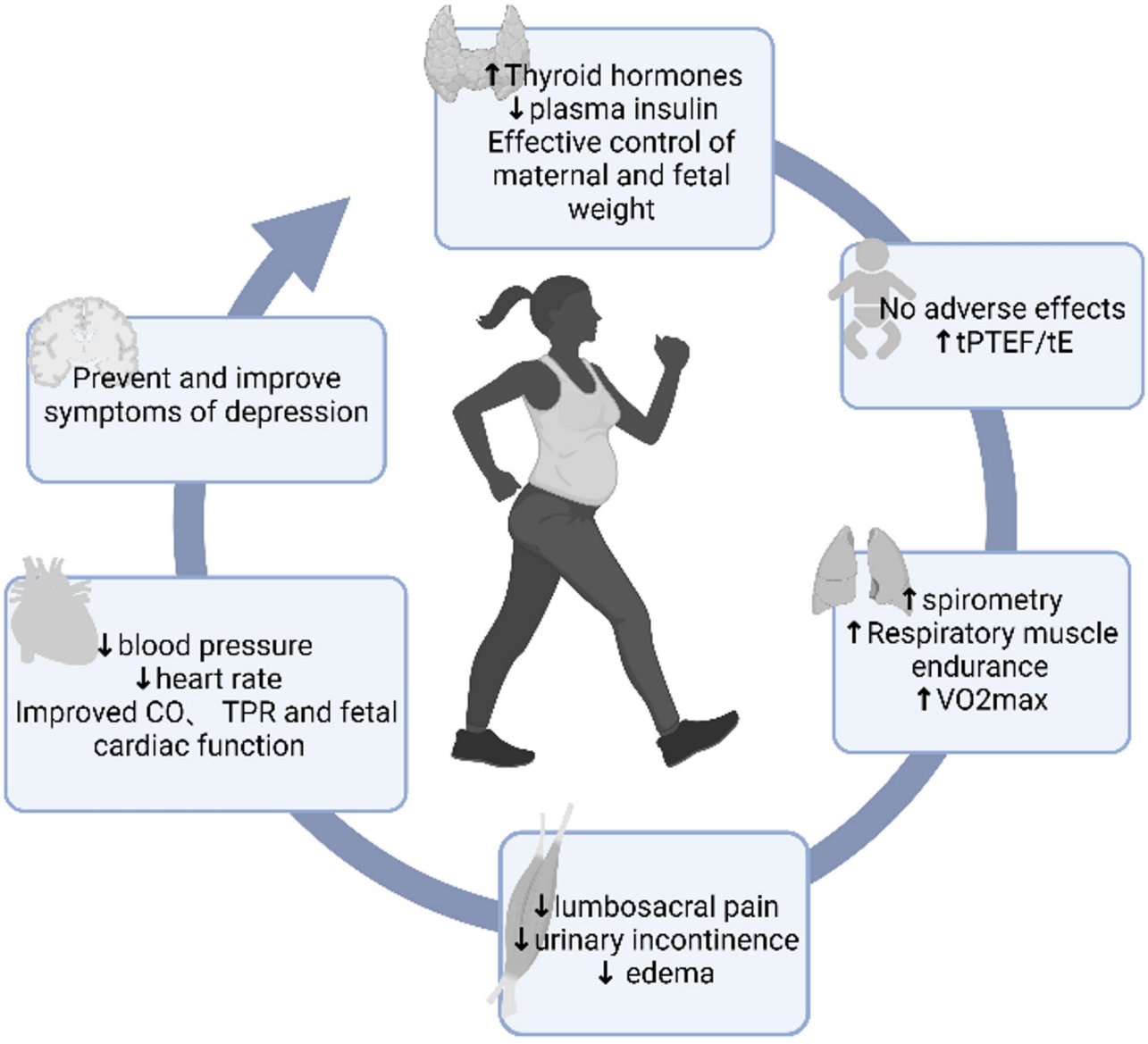

Fig. 2 Effect of exercise on maternal and fetal health

pregnancies; by assessing cardiac output (CO) and total peripheral resistance (TPR) via bioimpedance. After the 12-week intervention, mothers in the exercise group had lower systolic blood pressure and significant improvements in CO and TPR.

Endocrine and metabolic

Moderate exercise can regulate hormone secretion and improve metabolism. Studies have shown that exercise is associated with autonomic nervous system and endocrine activities, which can increase the levels of thyroid hormones, corticosterone, and endorphins; decrease the secretion of insulin, cortisol, brain serotonin (depression), etc., and control the body weight of mothers and fetuses [51–56]. Deep breathing during exercise stimulates the parasympathetic nervous system. Blood from Pregnant women undergoes postcirculatory oxygenation, which promotes the release of endorphins, which simultaneously inhibit the sympathetic nervous system and contribute to a reduction in cortisol release [57]. Exercise also reduces the risk of gestational diabetes in pregnant women and promotes glycemic control [58, 59]. Compared with nonexercising pregnant women, pregnant women who exercise exhibit lower insulin levels, increased insulin sensitivity, and improved glucose tolerance [51, 54, 60].

Respiratory system

Exercise during pregnancy improves maternal lung capacity and respiratory muscle endurance, improves

respiratory control and gas exchange, and effectively prolongs breath-holding exertion in favor of maternal preparation for natural childbirth [61, 62]. A recent meta-analysis indicated that prenatal exercise improved mothers' predicted/measured VO2max [58]. Aerobic exercise (e.g. swimming, jogging.) increases maternal cardiorespiratory endurance, improves oxygen uptake and utilization, and increases blood circulation and oxygen delivery to fetal and maternal tissues [61].

Musculoskeletal system

Low back and pelvic pain

In 2005, specific diagnostic criteria for pelvic girdle pain were used in a high-quality randomized controlled trial. In this study of 118 pregnant women with pelvic girdle pain, pelvic stabilization exercises neither reduced the intensity of pain nor shortened the recovery period after delivery [59]. This may be related to the inability to train the transverse abdominal muscles during pregnancy. As pregnancy progresses, physical activity has been found to mitigate the extent of biomechanical changes, such as reducing spinal loading, increasing joint stability, and contributing to better spinal alignment and segmental motion [63]. Compared to no exercise, prenatal exercise reduced the severity of low back pain, pelvic pain, or lumbopelvic pain during pregnancy and postpartum, although it did not reduce the risk of low back pain, pelvic pain, or lumbopelvic pain occurring at any point in time [63]. Exercises such as pelvic exercise programs, yoga, Pilates, slings, bobath balls, aerobic training, resistance training, and core stability training are often recommended for the treatment of low back pain and pelvic pain, and clinical Pilates was performed twice weekly for eight weeks for patients with low back pain during pregnancy and labor in the RCT trial by Sonmezer et al. [64]. Significant improvements in pain, disability, sleep, NHP physical activity subparameters, and lumbopelvic stability were found after Pilates exercise. The results of a systematic review and meta-analysis by Kazeminia et al. [65] suggested that pelvic floor muscle strengthening exercises significantly reduce low back pain. In addition, a meta-analysis showed that core stability exercises were more effective in reducing pain and may improve physical function in the short term in patients with chronic low back pain compared to general exercises [66, 67]. Therefore, these exercises could be considered part of a treatment plan for low back pain.

Pelvic floor muscle dysfunction

Exercise, particularly pelvic floor muscle training (PFMT), is an effective method for preventing and treating urinary incontinence during pregnancy; PFMT overcomes muscle wasting atrophy and improves muscle support and sphincter function to prevent and treat urinary incontinence [68]. Prenatal exercise involving PFMT was found to reduce the risk and symptom severity of prenatal and postnatal UI, and PFMT training with or without starting other types of exercise during pregnancy was associated with 50% and 37% reductions in the risk of developing prenatal and postnatal UI, respectively [69]. However, achieving that effect requires professional supervision and guidance.

Oedema In late pregnancy, the mother's body is prone to edema. It may lead to symptoms such as pain, heaviness, nocturnal cramps, and sensory abnormalities in women [28]. Exercises such as ankle pumps can improve edema by improving muscle pump function, increasing circulation, and reducing venous stagnation [70]. This correlation has been confirmed in numerous studies [71–74]. Clinical practice has also shown that ankle pump exercises, for example, can alleviate edema during pregnancy, with diuretic and edema-reducing effects in an RCT trial following 30 min of aerobic exercise per session for edema in women aged 20–33 weeks of gestation [75]. Recent studies have shown that the optimal frequency of ankle pumping exercises is 3–4 times/minute [76], but further research on specific protocols for other exercise interventions is needed.

Psychological

During pregnancy, mothers are most vulnerable in terms of their psychological balance and mood. Depression during pregnancy increases the risk of altered brain development, preterm labor, or intrauterine growth restriction [77, 78], and interventions are necessary to avoid negative effects on the fetus and mother. Medication is the most common treatment for patients with depression, although most antidepressants have been shown to have adverse effects on fetuses in animals [79]. However, most pregnant women are reluctant to accept medication due to the potential side effects of antidepressants on both the mother and the fetus [80]. Theoretically, the neurological changes associated with exercise birth, increased secretion of endorphins, serotonin, dopamine, norepinephrine, etc., can produce positive moods and feelings of well-being [77, 78, 81]. This correlation has been confirmed in numerous studies [82–85]. Other studies have shown that exercise during pregnancy improves symptoms associated with depression [86, 87]. In addition, in a recent RCT, Vargas-Terrones et al. [88] reported a reduction in the prevalence of late pregnancy and postpartum depression after a pregnancy exercise program of 60 min, 3 days per week, for 3 months. Similarly, several meta-analyses have indicated that exercise during pregnancy or prepregnancy has a preventive effect on depression and reduces the severity of depression [87, 89].

Effect of exercise on pregnancy outcomes

Examining pregnancy outcomes is the most reliable way of determining the success of pregnancy and labor. The following are the most critical pregnancy outcomes.

Fetal age at delivery

Sufficient time is required for the maturation of fetal organs and systems. Infants born prematurely are at risk of major health challenges throughout their lives, including chronic respiratory disease, neurodevelopment, and cardiovascular disease [90–92]. Excessive gestational age increases the risk of macrosomia, which is detrimental to both mothers and babies during labor [93]. Traditionally, it is believed that exercise increases the risk of miscarriage or preterm labor by decreasing the amount of circulation in the placenta; however, studies have shown that pregnant women who are sedentary during pregnancy are more likely to deliver prematurely [94], that there is no link between prenatal exercise and a shorter gestational age or an increased risk of preterm labor, and that exercise may even reduce the risk of preterm labor [95–97].

Mode of delivery

Different modes of delivery will have different effects on maternal and fetal health. Compared to vaginal delivery, cesarean section and instrument-assisted birth can produce sequelae such as scarring, chronic pain, pelvic adhesions, prolonged postpartum recovery time, or even irreversible damage [93, 98–100]. It has been demonstrated that exercise can counteract or mitigate the effects of pregnancy, e.g. physical activity during pregnancy improves the incidence of spontaneous delivery and neonatal Apgar scores [101].

Duration of labor

The duration of labor is closely related to the health of the mother and the fetus. Prolonged labor is a clinical problem in modern midwifery and can cause many problems for both mothers and babies [102] such as maternal fatigue, induction of labor, cesarean section and instrumental delivery, weak uterine contractions, maternal mortality and increased fetal distress, hypoxia, low Apgar scores, and ultimately fetal death [103]. However, a 2013 RCT stated that exercise did not affect labor. However, a recent RCT of twice-weekly Pilates exercise for 8 weeks in pregnant women at 26–28 weeks of gestation and delivery significantly reduced the length of the second stage of labor and increased maternal satisfaction with the labor process without complications for mothers and babies compared to those who did not exercise [104].

Conclusion

A review of existing studies indicates that physical activity and exercise during pregnancy are generally associated with minimal risks and that moderate-intensity exercise may offer benefits for certain maternal and neonatal outcomes. Current evidence suggests that encouraging pregnant women without contraindications to engage in moderate exercise could be a beneficial public health strategy; however, any exercise regimen should be adapted in consideration of normal anatomical and physiological changes during pregnancy and fetal well-being. We have further emphasized in the Discussion the necessity for future primary studies to clearly distinguish between different types of exercise and to analyze them as independent variables, as this will be critical for developing more precise exercise recommendations. It is important to note that the available evidence varies in quality, and many studies exhibit heterogeneity in exercise protocols, populations, and outcome measures. Furthermore, the long-term effects of prenatal exercise on offspring health remain an area requiring more robust investigation. Therefore, further high-quality, well-designed research is needed to determine the optimal type, frequency, duration, and intensity of exercise programs for pregnant women.

Limitations of study

A further limitation concerns the scope of included interventions. Our review focused primarily on low-to-moderate intensity exercise, which constitutes the majority of current clinical recommendations. We consciously excluded studies investigating high-intensity exercise regimens due to the highly controversial nature of the existing evidence and the significant heterogeneity in definitions and safety outcomes across studies. This decision, while necessary to maintain a clear and consistent focus, precludes a comprehensive synthesis of the potential benefits and risks associated with high-intensity exercise during pregnancy. This remains a critical area for future systematic reviews once a more robust and consistent evidence base emerges.

Abbreviations

CO	Cardiac Output
TPR	Total Peripheral Resistance
PFMT	Pelvic Floor Muscle Training
UI	Urinary Incontinence

Authors' contributions

YS wrote the manuscript, and all other authors (YW and PH) critically revised the drafts of the manuscript and approved the final version of the manuscript. YW and PH supervised all tasks.

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Data availability

This manuscript does not report data generation or analysis.

Declarations

Ethics approval and consent to participate
The type of article is a review and, therefore, there is no need for ethical approval and consent to participate.

Consent for publication
It is not applicable since there are no details, images, or videos relating to a person.

Competing interests
The authors declare no competing interests.

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